SM Graded Quiz M2

Question 1

In the Lewis (2001) formula, what role does the characteristic function $\phi(u)$ play?

- It defines the option maturity
- It encodes the distribution of the underlying asset process
- It calculates the risk-free rate
- It determines the strike price

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Question 2

Why is model calibration important in derivative pricing?

- It predicts future stock prices
- It determines the strike price of options
- It calculates historical volatility
- It finds parameter values that make model prices match current market prices

Question 3

If you needed to create a new option pricing method using Fourier transforms, what would be the essential requirement?

- Historical price data only
- A closed-form Black-Scholes solution
- The exact probability density function
- Knowledge of the characteristic function of the underlying process

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Question 4

What is the practical implication of Parseval's relation in Fourier-based option pricing?

- It determines the option's strike price
- It eliminates numerical errors
- It converts American options to European
- It allows computation of integrals in either time or frequency domain

Question 5

What mathematical theorem do we employ in analytically solving the CF integral for Black-Scholes?

- Cauchy's integral theorem
- Central limit theorem
- Mean value theorem
- Bayes' theorem

Question 6

In the expression $\phi_X(u) = \int e^{iux} f(x) dx$, what does $f(x)$ represent?

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- The cumulative distribution function
- The characteristic function of X
- The probability density function of X
- The moment generating function

Question 7

How does the characteristic function help when working with sums of independent random variables?

- The CF of the sum is the product of individual CFs
- The CF converts sums to single variables
- The CF of the sum is the sum of individual CFs
- The CF eliminates the need to consider independence

Question 8

What is the main computational benefit of having a closed-form characteristic function?

- It removes model risk
- It guarantees exact option prices

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- It eliminates all numerical methods
- It enables fast numerical integration for option pricing

Question 9

In the FFT implementation, what does the parameter N typically represent?

- The strike price
- The number of simulations
- The number of time steps
- The number of grid points for discretization

Question 10

If $N = 4096$ and the factor $g = 1$, how many grid points are used in the FFT?

- 1024
- 8192
- 2048
- 4096

Question 11

What is the main benefit of FFT when pricing

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options across many strikes?

- FFT computes prices for all strikes in a single computation
- FFT handles American features better
- FFT eliminates all numerical errors
- FFT is more accurate than other methods

Question 12

Why does FFT pricing work well for Black-Scholes even though we have a closed-form solution?

- FFT is always preferable to closed-form
- It validates the FFT method and serves as a benchmark
- FFT handles dividends better
- The closed-form is less accurate

Question 13

For high volatility $\sigma = 0.5$ with $S_0 = 100$, $K = 100$, $r = 0.05$, $T = 1$, what is the call option price

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under Lewis (2001) method?

- 21.79
- 18.25
- 25.6
- 15.9

Question 14

For a short-term call option with $S_0 = 100$, $K = 100, r = 0.05$, $\sigma = 0.3, T = 0.25$, what is the option price under Lewis (2001) method?

- 4.85
- 6.58
- 8.42
- 3.25

Question 15

When comparing Lewis to Carr-Madan FFT, which is generally preferred?

- They cannot be compared
- Carr-Madan is always superior
- Lewis is often preferred despite similar computational times

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- Lewis only works for European options

Question 16

What is the impact of the .real operation in BS_integral?

- It converts to risk-neutral probability
- It extracts the real part of the complex number result
- It ensures the stock price is positive
- It calculates the present value

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